Sampling Distributions & Central Limit Theorem

ANSWER KEY

Section 1: The Central Limit Theorem (CLT)

1 What does the Central Limit Theorem state about the sampling distribution of the sample mean $\bar{x}$?

a. When sample size n is sufficiently large ($n \geq 30$), the sampling distribution of $\bar{x}$ is approximately normal, regardless of the population shape ✓

b. As n increases, the population becomes normally distributed

c. The sample standard deviation approaches the population standard deviation

d. Every random sample will have an identical mean

The CLT states that for sufficiently large n (conventionally $n \geq 30$), the sampling distribution of $\bar{x}$ approaches normality even if the underlying population is skewed or non-normal.

2 If a population is already normally distributed with mean μ and standard deviation σ, what is the shape of the sampling distribution of $\bar{x}$ for a small sample of $n = 5$?

a. Exactly normal ✓

b. Uniform

c. Skewed

d. Cannot be determined without $n \geq 30$

When the original population is normally distributed, the sampling distribution of $\bar{x}$ is normal for any sample size n.

3 Does the Central Limit Theorem apply to the shape of the individual sample data values?

a. No, it applies only to the distribution of sample statistics across repeated sampling, not the sample data itself ✓

b. Yes, individual observations become normal as n grows

c. Only if the data has no outliers

d. Yes, this is known as the Law of Large Numbers

The CLT describes the sampling distribution of the statistic $\bar{x}$, never the distribution of individual data points in a single sample.

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Section 2: Mean and Standard Error of the Sample Mean

4 A population has mean $\mu = 80$ and standard deviation $\sigma = 20$. For random samples of size $n = 25$, what are the mean and standard error of $\bar{x}$?

a. $\mu_{\bar{x}} = 80$, $\sigma_{\bar{x}} = 4$ ✓

b. $\mu_{\bar{x}} = 80$, $\sigma_{\bar{x}} = 20$

c. $\mu_{\bar{x}} = 16$, $\sigma_{\bar{x}} = 4$

d. $\mu_{\bar{x}} = 80$, $\sigma_{\bar{x}} = 0.8$

$\mu_{\bar{x}} = \mu = 80$. Standard error: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{20}{\sqrt{25}} = \frac{20}{5} = 4$.

5 To cut the standard error of the sample mean $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ in half, the sample size must be:

a. **Quadrupled ($4n$)** ✓

b. Doubled ($2n$)

c. Halved

d. Increased by 10

Because n is under the square root in the denominator, multiplying n by 4 divides the standard error by $\sqrt{4} = 2$.

6 When using the formula $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$, what condition must be satisfied if sampling without replacement?

a. **The 10% condition: sample size $n \leq 0.10N$ (population size)** ✓

b. The Large Counts Condition: $np \geq 10$

c. $n \geq 30$

d. The population must be symmetric

The 10% condition ($n \leq 0.10N$) ensures that sampling without replacement is close enough to independent that the standard error formula remains valid.

Section 3: Sampling Distribution of Sample Proportion p-hat

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7 In a large population, 40% of adults support a referendum ($p = 0.40$). For random samples of $n = 100$, what are the mean and standard deviation of $\hat{p}$?

a. $\mu_{\hat{p}} = 0.40$, $\sigma_{\hat{p}} = 0.049$ ✓

b. $\mu_{\hat{p}} = 40$, $\sigma_{\hat{p}} = 4.9$

c. $\mu_{\hat{p}} = 0.40$, $\sigma_{\hat{p}} = 0.24$

d. $\mu_{\hat{p}} = 0.60$, $\sigma_{\hat{p}} = 0.049$

$\mu_{\hat{p}} = p = 0.40$. $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.40(0.60)}{100}} = \sqrt{0.0024} \approx 0.049$.

8 Which conditions justify modeling the sampling distribution of $\hat{p}$ with a normal distribution?

a. $np \geq 10$ **and** $n(1-p) \geq 10$ ✓

b. $n \geq 30$

c. $n \leq 0.10N$

d. $p = 0.50$

The Large Counts Condition requires both expected successes $np \geq 10$ and expected failures $n(1-p) \geq 10$.

9 Why is the sample proportion $\hat{p}$ called an unbiased estimator of p?

a. **The mean of its sampling distribution equals the true population parameter** p ✓

b. It never makes a prediction error

c. Its standard deviation is zero

d. It is computed from a simple random sample

An estimator is unbiased when the expected value (mean of its sampling distribution) equals the true population parameter being estimated.

Section 4: Calculations with Sampling Distributions

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10 A tire manufacturer knows tire lifespan is normal with $\mu = 50000$ miles and $\sigma = 4000$ miles. What is the probability that an SRS of 16 tires has a mean lifespan under 48000 miles?

a. $P(Z < -2.0) \approx 0.0228$ ✓

b. $P(Z < -0.5) \approx 0.3085$

c. $P(Z > 2.0) \approx 0.9772$

d. 0.0500

$\sigma_{\bar{x}} = \frac{4000}{\sqrt{16}} = 1000$. $z = \frac{48000 - 50000}{1000} = \frac{-2000}{1000} = -2.0$. Tail area is $P(Z < -2.0) \approx 0.0228$.

11 How does increasing sample size affect the bias and variability of a sample statistic?

a. **Variability decreases, while bias remains unchanged** ✓

b. Both bias and variability decrease

c. Bias decreases, variability remains unchanged

d. Both remain unchanged

Increasing sample size reduces sampling variability (standard error narrows), but has no effect on bias (which stems from study design).

12 If X_1 and X_2 are independent sample means from two populations, what is the standard error of $(\bar{x}_1 - \bar{x}_2)$?

a. $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ ✓

b. $\frac{\sigma_1}{\sqrt{n_1}} - \frac{\sigma_2}{\sqrt{n_2}}$

c. $\sqrt{\frac{\sigma_1}{n_1} + \frac{\sigma_2}{n_2}}$

d. $\frac{\sigma_1 + \sigma_2}{\sqrt{n_1 + n_2}}$

Variances of independent sample means add: $\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\sigma_{\bar{x}_1}^2 + \sigma_{\bar{x}_2}^2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$.

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